

9.

Ans: D

The gravitational potential at the point near the stars is given by

$$\phi = -\frac{2GM}{r}$$

The kinetic energy of the rock is found using energy conservation:

$$KE_{\infty} + PE_{\infty} = KE_r + PE_r$$

$$0 + 0 = \frac{1}{2}mv^2 - \frac{2GMm}{r}$$

$$v = \sqrt{\frac{4GM}{r}}$$



